[16] Gradient Descent at the Edge of Stability

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Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability — RQ2, RQ4

Problem: Classical convergence arguments assume that a sufficiently small gradient-descent step leads to a steady decrease in loss. Do neural networks actually train in this small-step regime? **Method:** The authors train networks by *full-batch gradient descent*, measuring both loss and the top eigenvalue of the loss Hessian throughout training. **Main finding:** The top eigenvalue often rises to around $2/\eta$ and hovers just above it while short-term loss fluctuates yet decreases over longer periods. This is an *empirical observation*, not a general convergence theorem.

1. Why the threshold $2/\eta$ appears

Start from a one-dimensional quadratic loss about its minimum x^*:

$$L(x) = \tfrac{1}{2}a(x - x^*)^2, \qquad a > 0.$$

GD with step size η updates

$$x_{t+1} = x_t - \eta a(x_t - x^*).$$

Write the error $e_t = x_t - x^*$. Then

$$\boxed{e_{t+1} = (1 - \eta a)e_t, \quad e_t = (1 - \eta a)^t e_0}.$$

Errors shrink if $|1 - \eta a| < 1$, i.e. $0 < \eta a < 2$. If $\eta a > 2$, the error changes sign each iteration *and grows*: oscillatory instability. For a positive definite Hessian H, decompose the local error into Hessian eigenvectors. Each coordinate scales by $(1 - \eta\lambda_i)$, so the limiting sharpness is $\lambda_{\max}(H) \approx 2/\eta$.

What this calculation proves: A *fixed quadratic* becomes unstable when its largest eigenvalue exceeds $2/\eta$. It does *not* prove that nonlinear neural-network training diverges whenever the instantaneous Hessian crosses that threshold, because both the position and curvature change during training.

2. What Cohen et al. actually measured

The paper defines *sharpness* precisely as the top Hessian eigenvalue,

$$S(\theta_t) = \lambda_{\max}(\nabla^2 L(\theta_t)).$$

For constant-step full-batch GD, experiments across several architectures and tasks reveal a recurring trajectory: (i) **progressive sharpening**, when $S(\theta_t)$ increases; (ii) **edge of stability**, when $S(\theta_t)$ stays around or slightly above $2/\eta$ even as training continues. Short-run loss need not decrease at every step. The empirical pattern challenges the idea that training always follows the descent-lemma regime $\eta \le 1/\ell$.

3. Interpret the local dynamics correctly

Near a changing parameter point, a quadratic Taylor approximation gives

$$L(\theta_t + \delta) \approx L(\theta_t) + \nabla L(\theta_t)^\top \delta + \tfrac{1}{2}\delta^\top H_t \delta.$$

When ηS_t approaches 2, perturbations along the sharpest eigenvector oscillate strongly. But since H_t changes as the weights move, a period of instability may be followed by lower sharpness. *The paper documents this feedback but does not establish a complete theorem for the mechanism*; later self-stabilization studies pursue that explanation.

Connection to our survey. This paper motivates Section VII: practical full-batch GD can succeed outside the regime assumed in many smooth, monotonic-descent analyses. The relevant variables are not just the gradient magnitude but also evolving curvature and the chosen step size.

4. Assumptions, scope, and limits

The central evidence is experimental, with *full-batch* GD and prescribed step sizes. The authors explicitly caution against inferring an established link between their Hessian sharpness measure and generalization. The observed

edge should not be stated as a universal law of minibatch SGD or Adam; in fact the paper notes that the sharpness for SGD need not flatline at the same threshold. The $2/\eta$ boundary is exact for fixed positive quadratic curvature, heuristic as a moving local marker for nonlinear training. **Our takeaway.** The striking message is not that GD becomes unstable and fails, but that it can train successfully while repeatedly approaching a boundary where its local quadratic approximation would become unstable. **Source:** https://arxiv.org/pdf/2103.00065 (esp. pp. 1–3 and experimental sections).